

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Year: 2013

Subject: 392133 Physics III

Section: 15-16

Date: 1 August 2013

Time: 8.00-10.00

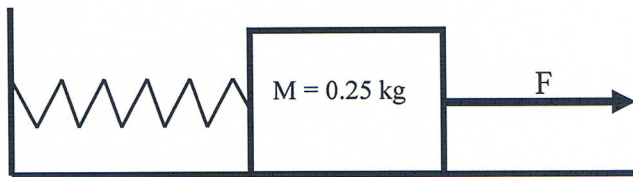
Name:.....ID:.....Class:.....

Instructions

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
 2. No documents are allowed to be taken inside the examination room.
 3. Textbook and dictionary are **NOT** allowed.
 4. A scientific calculator is allowed.
 5. Express your answer in English. Other languages will be discarded.
 6. Electronic communication devices are **NOT** allowed in the examination room.
 7. Examination has 7 pages (including this page), 6 problems and total scores of 60 points.
 8. Write solutions and answers on the examination sheet.
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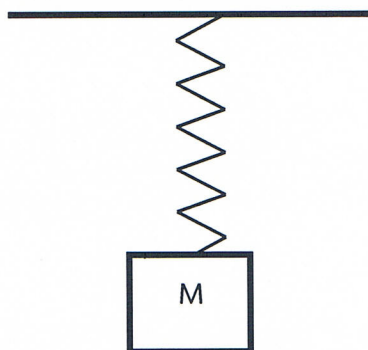
1. A 0.250 kg block resting on a frictionless, horizontal surface is attached to a spring having force constant 83.8 N/m as shown in the figure. A horizontal force F causes the spring to stretch a distance of 5.46 cm from its equilibrium position.

- [1 point] What is the restoring force due to the spring when the spring is stretched?
- [1 point] Find the value of F .
- [2 points] What is the total energy stored in the system when the spring is stretched?
- [2 points] Find the magnitude of the acceleration of the block immediately after the applied force is removed.
- [2 points] Find the speed of the block when it first reaches the equilibrium position.
- [2 points] If the surface is not frictionless but the block can move along the surface only one roundtrip. Determine the kinetic friction coefficient (μ) between the block and the surface.



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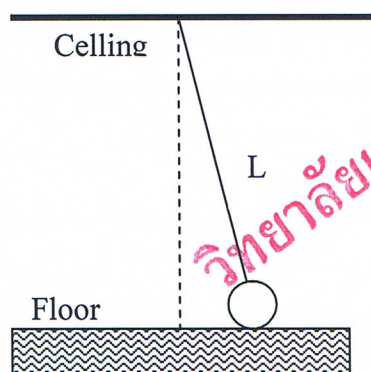
2.1 [4 points] A vertical spring stretches 3.9 cm when a 10 g object is hung from it. The object is replaced with a block of mass 25 g that oscillates up and down in simple harmonic motion. Calculate the period of motion.



2.2 A man enters a tall tower, needing to know its height. He notes that a long pendulum extends from the ceiling almost to the floor and that its period is 12.0 s.

(a) [2 points] How tall is the tower?

(b) [2 points] If this pendulum is taken to the Moon, where the free-fall acceleration is 1.67 m/s^2 , what is its period there?



2.3 [2 points] The angular displacement of a pendulum is represented by the equation

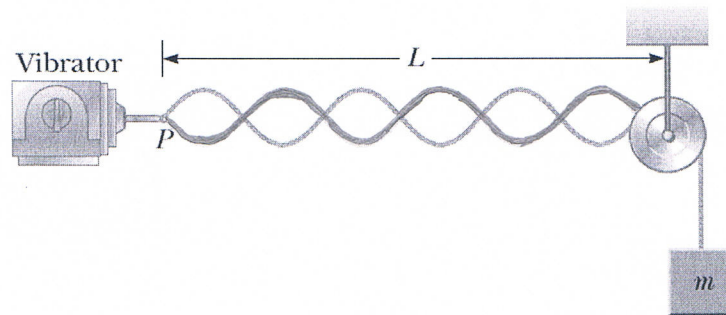
$$\theta = 0.320 \cos \omega t$$

where θ is in radians and $\omega = 4.43 \text{ rad/s}$. Determine the period and frequency of the pendulum.

4. In the arrangement shown in the figure, an object of mass $m = 5.0$ kg hangs from a cord around a light pulley. The length of the cord between point P and the pulley is $L = 2$ m.

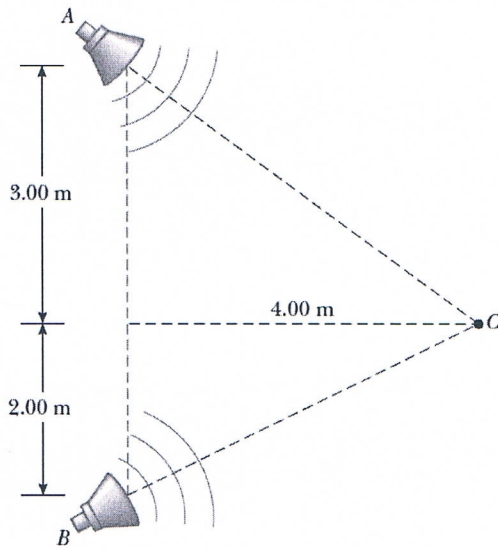
(a) [5 points] When the vibrator is set to a frequency of 150 Hz, a standing wave with six loops is formed. What must be the mass of the cord?

(b) [5 points] How many loops (if any) will result if m is changed to 10 kg?

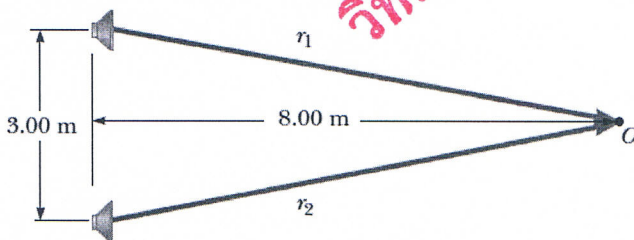


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5.1 [5 points] Two small speakers emit sound waves of different frequencies. Speaker A has an output of 1.00 mW, and speaker B has an output of 1.50 mW. Determine the sound level (in dB) at point C if both speakers emit sound.



5.2 [5 points] Two loudspeakers are placed above and below each other, as in the figure and driven by the same source at a frequency of 4.50×10^2 Hz. An observer is in front of the speakers (to the right) at point O, at the same distance from each speaker. What minimum vertical distance upward should the bottom speaker be moved to create destructive interference at point O? The temperature at 20 °C is assumed.



6.1[5 points] Standing at a crosswalk, you hear a frequency of 560 Hz from the siren of an approaching ambulance. After the ambulance passes, the observed frequency of the siren is 480 Hz. Determine the ambulance's speed from these observations. The temperature at 20 °C is assumed.

6.2 [5 points] When an open metal pipe is cut into two pieces, the lowest resonance frequency for the air column in one piece is 256 Hz and that for the other is 440 Hz. The temperature at 20 °C is assumed.

- (a) What resonant frequency would have been produced by the original length of pipe?
- (b) How long was the original pipe?

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3.1 [4 points] Transverse waves travel with a speed of 20.0 m/s in a string under a tension of 6.00 N. What tension is required for a wave speed of 30.0 m/s in the same string?

3.2 [6 points] There are two rectangular wave pulses on a stretched string traveling toward each other. Each pulse is traveling with a speed of 1 cm/s and has the height and width shown in the figure. If the leading edges of the pulses are 8 cm apart at $t=0$, please sketch the shape of the string at $t = 4$ s, and $t = 5$ s.

